

AFSANA RAHMAN MOU

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Permanent Resident ◇ Sponsorship not required

SUMMARY

I am currently a data science apprentice at Sequaretech Inc., specializing in User and Entity Behavior Analytics (UEBA). My work involves developing a customized LLM log parser with open-source LLM models such as Flan-T5, Langchain. While working at the university my research focused on optimizing deep learning models through compression techniques like pruning and quantization for audio classification. I also explored deploying these models on edge devices, such as the Raspberry Pi and Nvidia Jetson Nano. In addition to my research, I serve as an NVIDIA University Ambassador and am a certified instructor for NVIDIA DLI's Deep Learning workshops.

EDUCATION

Ph.D. in Computer and Information Sciences(Concentration in Computer Science)	December 2023
Thesis : Evaluating Performance of Deep Learning Model Compression Techniques on Edge Devices for Audio.	
University of Arkansas at Little Rock, Arkansas	GPA 3.75/4
Master of Science in Computer Science ,University of Arkansas at Little Rock, Arkansas	December 2022
Bachelor of Science in Electrical and Electronic Engineering	December 2018
Chittagong University of Engineering & Technology, Bangladesh	

MACHINE LEARNING SKILLS

Tools & Packages	TensorFlow, PyTorch, LLaMA 3, Vector DataBase, Flan-T5, Mistral 7B, NLP, Transformers, Diffusion Model, RAG, Random Forest, LLM, Prompt Engineering.
Programming Languages	Python, C/C++, MATLAB, R, SQL, Tableau, AWS, GCP, SAS, PySpark.

EXPERIENCE

Data Scientist, Sequaretek Inc, Arkansas	July 2024 - Present
• Build an LLM log parser that enables accurate, context-aware extraction and interpretation of log data, enhancing automation, anomaly detection, and troubleshooting efficiency. • Create an LSTM model to analyze email exfiltration for signs of unauthorized data transfer. • Developing Robust Random Cut Forest algorithms for User and Entity Behavior Analytics(UEBA). • Building a chatbot utilizing LLaMA 3, RAG, LangChain, and other generative AI techniques.	
Teaching Assistant, NSA Cybersecurity Workforce Certificate Program	Feb 2024 - Aug 2024
• Create modules for Digital forensics and Post-Quantum Cryptography and taught students about quantum computing, IBM Q experience,data storage, Encryption, investigation types, metadata. • Took labs on Cognitive Computing and Robotic Process Automation Analysis and lessons include building medical chatbot, digital summerization, and speech recognition and diarization.	
Graduate Assistant, University of Arkansas at Little Rock, Arkansas	Jan 2021 - Dec 2023
• Use a Large Language Model (LLM) to answer questions based on provided text. • Optimized deep learning models through compression techniques like pruning and quantization for audio classification and deploying these models on edge devices and achieved 86% accuracy. • Worked on speech emotion analysis to classify emotional states in audio samples from the TESS and RAVDESS datasets. Used MFCC for feature extraction and achieved 98% accuracy with LSTM.	

- Use pre-trained, modern LLMs to solve various NLP tasks such as token classification, text classification, summarization, and question answering .

Instructor, NVIDIA Deep Learning Institute

Jan 2021 - Present

- Worked as Lead Instructor in more than 15 Workshops on NVIDIA DLI Fundamentals of Deep Learning.
- Achieved NSF EPSCoR SEED GRANT to teach people about deep learning.

PROJECTS

- **Deep Learning Model Compression on Edge Devices(NVIDIA Jetson Nano, Raspberry Pi)**
Compared Autoencoder and Transformer models for audio classification using MFCC and Mel Spectrograms. Applied pruning, quantization, and knowledge distillation for edge device deployment and optimized hyperparameters to achieve peak model performance.
- **Active Learning Monitoring in Classroom**
Analyzed the performance of Convolutional Neural Networks and LSTM models on a custom dataset, classifying instructor recordings into voice categories using Mel spectrograms. Explored hyperparameter impacts such as window size, hop length, and nfft. (NSF Project)
- **Handwritten Image Generation using Diffusion Model**
Created a diffusion model using U-Net architecture to generate handwritten images based on the MNIST dataset. Optimized the model using Group Normalization, GELU, Rearrange Pooling, and Sinusoidal Position Embeddings, achieving more than 95% accuracy.
- **Census Bureau Name Entity Recognition**
Implemented an information integration methodology to enhance entity resolution and record linkage quality. Focused on refining name and address standardization and optimizing record linkage through group structure considerations. (NSF Project)

PUBLICATIONS

- Performance Analysis of Deep Learning Model-Compression Techniques for Audio on Edge Devices, ([link](#)).
- Intelligent Traffic Control System Using YOLO Algorithm for Traffic Congested Cities, ([link](#)).
- Detection, reduction and filtration of cancer cells through a new DNA polymerization sequence approach, ([link](#)).
- Active Learning Monitoring in Classroom using Deep Learning Frameworks, CANADA ([link](#)).
- Deep Learning Approaches for Classroom Audio Classification using Mel Spectrograms,Bulgaria([link](#)).

CERTIFICATIONS

- CISCO *Endpoint Security* , *Cyber Threat Management*, *Network Support and Security*
- CISCO *Introduction to Cybersecurity*, *Ethical Hacker*
- NVIDIA DLI Certificate – Generative AI with Diffusion Models. ([link](#))
- NVIDIA DLI Certificate – Getting Started with AI on Jetson Nano. ([link](#))
- NVIDIA DLI Certificate – Building Transformer-Based Natural Language Processing Applications. ([link](#))

AWARDS & EXTRA-CURRICULAR ACTIVITIES

- CSTEM Continuing Success Award for Excellent Academic Result, University of Arkansas, Little Rock.
- NVIDIA DLI University Ambassador ([link](#)), Member at IEEE & Women In Engineering.